

UAV-Assisted MEC System With Mobile Ground Terminals: DRL-Based Joint Terminal Scheduling and UAV 3D Trajectory Design

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Abstract—In this paper, we investigate an unmanned aerial vehicle (UAV)-assisted edge computing network in a sophisticated three-dimensional (3D) scenario, e.g., a post-disaster urban area with multiple mobile ground terminals (GTs) for resilience. The mission of the UAV is to first collect a set of computation tasks from multiple GT sources, then process them and make joint decisions (computation results) which are finally transmitted to another set of GT destinations. An efficient design is provided aiming at minimizing the total time cost including costs for the task offloading, computations, and decision transmissions, via jointly designing the UAV's trajectory and the scheduling of communications from and to different GTs. In particular, we take into account the practical assumptions on the GTs' mobility, on the obstacle avoidance to 3D buildings, and on that the UAV is possible to fly among buildings, i.e., the flying altitude can be lower than the buildings' heights. To address the established complex and non-convex problem, we first provide a problem transformation by exploiting Markov decision process (MDP), and then propose a deep reinforcement learning (DRL) approach-based solution with a multi-step dueling DDQN (D3QN) method. As a result, an efficient solution is achieved, where the UAV acts as the agent to continuously explore and improve its mobile strategy during interacting with the environment under the proposed method. The simulation results show the advantages of the proposed 3D trajectory design in comparison to a 2D one. In addition, the results confirm the robustness of the proposed design in terms of different GTs' mobility models and UAV flying height/area limits, and illustrate the superiority of unrestricted feasible domains for the UAV flight.

Index Terms—3D trajectory design, deep reinforcement learning, obstacle avoidance, probabilistic LoS channel, UAV.

I. INTRODUCTION

IN RECENT years, unmanned aerial vehicles (UAVs) have been considered an efficient way to improve the performance of ground wireless networks [1], [2], due to their flexible mobility [3], fast deployment [4], as well as the high probability of line-of-sight (LoS) channel [5]. These attractive benefits have recently sparked a rapidly expanding enthusiasm in both academia and industry, leading to the development of new applications. For example, the UAV can be deployed to perform data collection and dissemination from ground terminals (GTs) [6] or as a flying relay [7]. In addition, the UAV also could serve as a flying mobile edge computing (MEC) node to help the GTs compute tasks [8]. In particular, as the flying MEC node, the UAV can carry MEC equipment in proximity to GTs to provide both the computation and the communication services, so as to reduce energy consumption, task completion time, and transmission delay.

For a UAV-assisted MEC system, existing studies have shown that by determining an appropriate deployment location, the system performances can be improved, e.g., in terms of MEC service coverage [9], offloading capability [10], and energy efficiency [11]. Moreover, via efficient trajectory (UAV's time-varying location) designs such performances are further boosted. On the one hand, the high mobility of UAVs, through their trajectory design, enables the MEC network to serve GTs in critical scenarios (mountainous areas and post-disaster). On the other hand, via efficient trajectory designs UAVs and GTs are able to establish high-probability LoS channels which result in high-speed offloading and decision transmissions. A set of recent works have confirmed the significant gain of UAV trajectory design for the UAV-assisted MEC systems [12], [13], [14]. The authors in [12] provide a UAV trajectory design aiming to minimize the overall energy consumption during the task offloading. Also to minimize the total energy consumption of the system, the authors of [13] proposed a three-stage iterative algorithm to jointly optimize the UAV's beamforming vectors, the UAV's central processing unit (CPU) frequency, and the UAV's trajectory. In addition, the work in [14] minimizes the system's total power consumption by jointly designing

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the UAV trajectory and resource allocation. Furthermore, with the rapid development of the machine learning (ML) industry, a set of ML-based trajectory designs have been provided for UAV-assisted wireless networks, e.g., authors in [15] introduce a deep reinforcement learning (DRL) based approach to minimize the UAV mission completion time by jointly optimizing the UAV's trajectory and the association between UAV and ground base stations, while a semi-distributed resource management approach is proposed in [16] to minimize the system's sum power also for UAV-assisted MEC systems.

The above trajectory design works [9], [10], [11], [12], [13], [14], [15], [16] are conducted under the assumptions of a simplified LoS-dominant channel model and a two-dimensional (2D) trajectory, i.e., the UAV flies at a fixed altitude. In fact, via enjoying the freedom of flying altitude, a 3D trajectory design is expected to achieve a higher performance in comparison to a 2D one. In particular, an efficient design of UAV's 3D trajectory also faces two significant challenges. On the one hand, optimizing a 3D trajectory requires a simultaneous consideration of the UAV's motion in both horizontal and vertical directions, which leads to a significantly high complexity in comparison to a 2D case. On the other hand, the 3D trajectory is restricted more by the detailed characteristics of the environment, which requires the design to carefully take into consideration environmental factors such as altitude variation, multi-dimensional propagation characteristics, and the effects of obstacles and blockages on the channel. Hence, for a UAV-assisted multi-GT MEC network, it is essential to take the practical probabilistic LoS channel states into account.

Under a practical probabilistic LoS channel model, the authors in [17], propose a UAV-assisted multi-GT framework by optimizing the 3D UAV trajectory to maximize the minimum (average) expected rate among GTs. The authors of [18] provide a 3D trajectory design method for harvesting solar energy and collecting data from multiple smart GTs. In addition, the works of [19], [20] exploit multiple UAVs to assist multiple GTs and design the UAV's 3D trajectory to enhance the data collection performance. Moreover, the works of [21], [22], [23] enhance the communication performance of the UAV as a mobile MEC node by optimizing its 3D trajectory. However, the aforementioned works [12], [13], [16], [17], [18], [19], [20], [21], [22], [23] have the assumption that GTs are fixed, i.e., the mobility of GTs is ignored. It is worth noting that the GTs may be in a mobile state under a practical scenario leading to that the mobility of GTs cannot be ignored. Furthermore, the above works [12], [13], [15], [16], [17], [18], [19], [20], [21], [22], [23] are conducted under the assumption that the UAV is flying over buildings, without considering the case of low-altitude flying of the UAV,¹ and thus ignoring the UAV's obstacle avoidance issue, which usually results in a challenging problem. However, in certain

urgent or military applications where the UAV has to fly at a low altitude among buildings, the obstacle avoidance becomes an issue that has to be addressed.

Recently, the mobility of GTs is modeled as a random behavior in recent studies when providing ML-based UAV trajectory designs [24], [25], while the obstacle avoidance to 3D buildings is ignored and it only focuses on the current return. To the best of our knowledge, for practical UAV-assisted MEC networks with multiple mobile GTs, considering the 3D feasible UAV flying area with obstacle avoidance, efficient joint designs on UAV 3D trajectory together with the communications scheduling (including the MEC tasks offloading and the decision transmissions) are still open issues. Compared with the existing works [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], we consider a UAV-enabled MEC scenario with more practical assumptions, e.g., with respect to the mobility of GTs, feasible UAV flying zones amidst buildings, obstacle avoidance in a 3D environment, and the utilization of a probabilistic LoS channel for communication with the 3D buildings. In the considered scenario, the UAV assists multiple mobile GTs to complete the tasks of data offloading, calculation, and decision transmission. Assuming that all GTs are moving at all times and that the UAV can fly between buildings (under obstacle avoidance constraints), we aim at minimizing the whole operation time cost via joint determining the 3D trajectory and the communication schedule from and to different GTs. The formulated problem is non-convex and difficult to solve the dynamic environment caused by the mobility of GTs, i.e., being intractable via applying analytical approaches (such as convex optimization or matching theory). In addition, compared to 2D trajectory design, the expanded exploration space of 3D trajectory design results in slower convergence of the algorithm. Consequently, it is essential to design algorithms that accelerate convergence and enhance overall performance. To solve such issues, we provide a joint design based on a modified reinforcement learning (RL) algorithm, i.e., the UAV acts as an agent acquiring the information of GTs in the process of interacting with the environment, learns the underlying model of the GTs' random moving behavior (through continual interaction learning between the agent and the environment), and adjusts its trajectory accordingly. The main contributions of this paper are presented as follows:

- *Design under Practical Assumptions:* Taking into account the practical assumptions of the mobility of GTs, obstacle avoidance to 3D buildings, feasible UAV flying zone among buildings, and probabilistic LoS channel model (due to the signal blockage from buildings), we provide a joint design on the 3D UAV trajectory and the GTs' communication scheduling to minimize the whole operation time.
- *Modified Learning-based Joint Design Solution:* To solve the formulated non-convex problem which is actually analytically intractable due to the random mobility of GTs and a large number of coupled variables, an efficient design approach for optimizing the trajectory of the UAV and the communication scheduling of GTs is proposed, including a modified learning-based method for the UAV's trajectory designing and a greedy scheme for mobile GTs' scheduling. In particular, we first transform the established

¹It is known that in several countries and cities, urban policies and safety regulations have limited the flight height of UAV to be sufficiently high than the buildings. On the other hand, it should be pointed out that it is still essential to consider in the design the possibility of UAV flying at a low altitude in emergency scenarios, i.e., military applications, emergency rescue, etc., and provide efficient designs accordingly. In such scenarios where the UAV could fly lower than the buildings, i.e. the UAV can approach closer to the GTs/users, the transmission performance of the entire system can be significantly boosted.

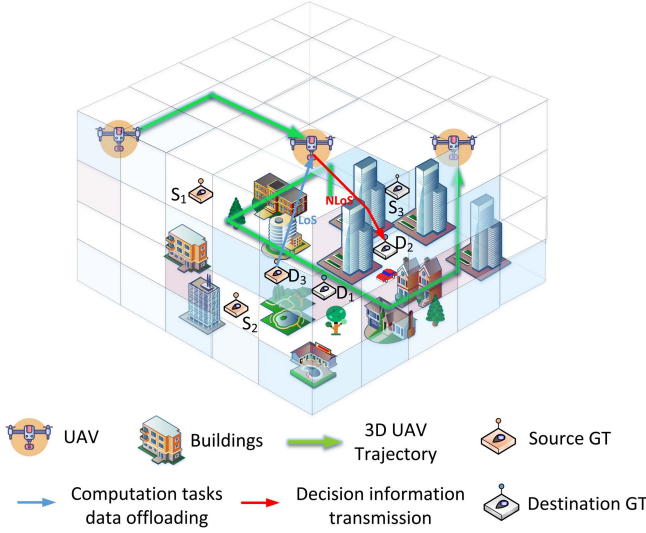


Fig. 1. UAV-assisted multi-ground terminals data operation system in probabilistic LoS channel.

problem into an MDP problem. Subsequently, we provide a modified DDQN algorithm with a dueling network and multi-step bootstrapping technology, namely the multi-step dueling DDQN (D3QN) method. Compared with the traditional DDQN method [24], [25], the proposed method with dueling network boosts the performance of the system, and multi-step bootstrapping technology with a suitable step makes the algorithm converge faster. Following the proposed D3QN method, an efficient DRL-based solution to the joint design problem is provided.

- *Simulative Evaluations:* The simulation results show the superiority of the proposed design. The 3D feasible UAV flying area with buildings avoidance can yield additional performance benefits. In addition, the performance improvement in 3D trajectory design, compared to 2D trajectory design, is primarily attributed to its increased flexibility in balancing angle and transmission distance. Moreover, the robustness of the proposed design has been confirmed, i.e., with respect to the moving modes of GTs, and the UAV flying height/area limits.

The rest of this paper is organized as follows. Section II introduces the preliminary. Section III formulates the problem. In Section IV, we propose a DRL-based 3D trajectory design. In Section V, the simulation results are presented. Finally, the work is concluded in Section VI.

II. PRELIMINARY

In this section, we first describe the considered UAV-assisted MEC system. Then we introduce the channel and computational models and finally present the obstacle avoidance model.

A. System Description

We consider a UAV-assisted MEC network with one UAV and massive mobile GTs in a sophisticated 3D urban environment, e.g., a post-disaster urban area, as shown in Fig. 1. The UAV

serves as a flying MEC node to assist mobile GTs in data collection, computation, and decision-making tasks while also requiring obstacle avoidance during flight near buildings. The GTs are randomly distributed in a certain ground geographical area $\mathcal{D}$. Among them, there are a set of I source GTs, denoted by $\mathcal{I} = \{S_1, \dots, S_I\}$, and a set of J destination GTs, denoted by $\mathcal{J} = \{D_1, \dots, D_J\}$. In particular, set $\mathcal{I}$ and set $\mathcal{J}$ could have a non-empty intersection, a GT can serve as both source GT and destination GT. The time-varying locations of source GT $i \in \mathcal{I}$ and destination GT $j \in \mathcal{J}$ at time instant t are denoted by $\mathbf{w}_i(t) = [x_i(t), y_i(t), 0]$ and $\mathbf{w}_j(t) = [x_j(t), y_j(t), 0]$, respectively. Additionally, the UAV's time-varying 3D location is denoted as $\mathbf{q}(t) = [x(t), y(t), z(t)]$ with an arbitrary initial location expressed as $\mathbf{q}(0) = [x(0), y(0), z(0)]$. Notably, the mobility of the GTs affects the performance of UAV trajectory design, as the GTs are assumed to move while offloading data tasks or receiving decision information. This assumption is practical and applicable in scenarios such as road inspection, detection, and photography.

The whole operation time of the considered network, denoted by T , is divided into two stages, as shown in Fig. 2 on the next page. During the first stage, the UAV collects data on computation tasks from the source GTs, and then processes these tasks and makes joint decisions. Subsequently, these data packets of decision information are transmitted to their corresponding destination GTs in the second stage. For ease of discussion, we uniformly discretize the whole operation time T into N time slots with the same length of σ and let $\mathcal{N} \triangleq \{0, 1, \dots, N\}$ denote the set of discrete time slots. Note that σ should be small enough that within each time step.² Thus, the distance between the UAV and GTs can be treated as approximately unchanged, or the channel gain towards the UAV or towards GTs, remains approximately constant. As such, the location coordinates of the UAV, the i -th source GT and the j -th destination GT can be re-expressed as $\mathbf{q}_n = [x_n, y_n, z_n], \forall n \in \mathcal{N}$, $\mathbf{w}_{i,n} = [x_{i,n}, y_{i,n}, 0], \forall i \in \mathcal{I}, \forall n \in \mathcal{N}$, and $\mathbf{w}_{j,n} = [x_{j,n}, y_{j,n}, 0], \forall j \in \mathcal{J}, \forall n \in \mathcal{N}$, respectively.

Denote the data size that source GT i needs to offload for calculation by Ξ_i in bits, and the workload for processing the task by W_i in Mcycles. In addition, the data size of decision information for destination GT j is denoted by Ξ_j in bits. Moreover, the total number of task offloading time slots of source GT i is denoted by N_i^U , and the total number of offloading time slots is denoted by N^U , i.e., $N^U = \sum_{i=1}^I N_i^U$. Note that the UAV can process a task immediately after successfully decoding the data. Given a fixed UAV computation frequency, assume the time taken by the UAV to process data is N^C . The total number of time slots of the first stage is actually the sum of the time slots for offloading all tasks and the total processing time slots of the last offloaded task, i.e., $N^U + N^C = \sum_{i=1}^I N_i^U +$

²The smaller the value of σ is, the more accurate the obtained trajectory solution is to the optimal one, but the complexity of the algorithm will increase. In addition, during trajectory solving, there is typically an assumption that the location and channel state information of the UAV remain unchanged within a time slot. An excessively large value of σ will make this condition not met, resulting in imprecise trajectory solutions. Thus, the selection of σ shows a significant trade-off between the system performance and algorithmic complexity.

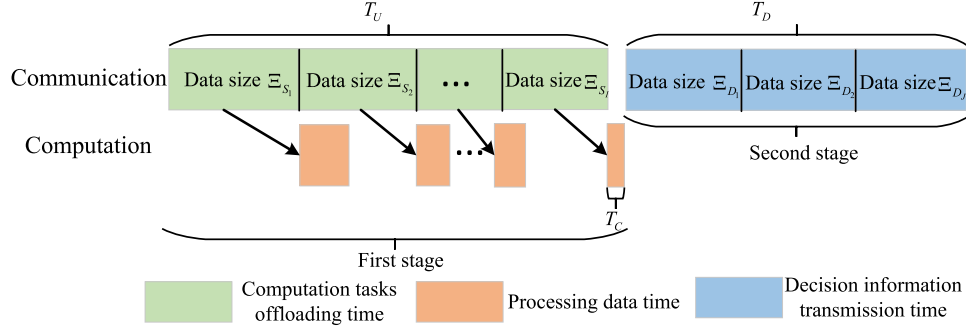


Fig. 2. Illustration of the whole UAV-assisted multi-ground terminals data operation process.

N^C , as shown in Fig. 2. Similarly, the total time slots of decision transmission N^D is expressed as $N^D = \sum_{j=1}^J N_j^D$, where N_j^D is the total time slots of sending the decision information to the destination GT j . Based on the above discussion, we have $N = N^U + N^C + N^D$ and $T = N\sigma$, where N denotes the whole operation time slots.

B. Probabilistic LoS Channel Model

In this paper, we adopt a statistically probabilistic LoS channel model [26] to determine the UAV-GT links, where the LoS and NLoS probabilities of a link are jointly considered. In such model, the elevation angle and the statistical parameters related to the environment are actually the main factors influencing the probability of geometrical LoS between the UAV and each GT [26]. Let $c_{i,n}$ and $c_{j,n}$ represent the channel state of source GT i and destination GT j with the UAV, where $c_{i,n} = 1$ and $c_{j,n} = 1$ indicate that the UAV-(source GT i) link and UAV-(destination GT j) link are in LoS state, while $c_{i,n} = 0$ and $c_{j,n} = 0$ represent that the UAV-(source GT i) link and UAV-(destination GT j) link are in NLoS state. According to [26], a commonly used probability LoS model is based on a simple logistic function in the form of

$$\mathbb{P}(c_{i,n} = 1) = \frac{1}{1 + ae^{-b(\theta_{i,n}-a)}}, \quad (1)$$

where a and b are parameters with constant values depending on the environment, e.g., the distribution and the heights of buildings. In addition, $\theta_{i,n}$ denotes the elevation angle between the UAV and source GT i in time slot n , which is given by

$$\theta_{i,n} = \frac{180}{\pi} \arctan \left(\frac{z_n}{\|\mathbf{q}_n^\lambda - \mathbf{w}_{i,n}^\lambda\|} \right), \quad (2)$$

where $\mathbf{q}_n^\lambda = [x_n, y_n]$ and $\mathbf{w}_{j,n}^\lambda = [x_{j,n}, y_{j,n}]$ are the horizontal coordinates of the UAV and destination GT j , respectively. To enhance the degree of preciseness of the probabilistic LoS channel model, the LoS probability can be approximately represented by the generalized logic function as [26]

$$P_{i,n}^L \triangleq \mathbb{P}(c_{i,n} = 1) = B_3 + \frac{B_4}{1 + e^{-(B_1+B_2\theta_{i,n})}}, \quad (3)$$

where $B_1 < 0$, $B_2 > 0$, $B_3 > 0$, and $B_4 = 1 - B_3$ are constants determined by the environments. So the corresponding NLoS

probability between the UAV and source GT i can be represented as

$$P_{i,n}^N \triangleq \mathbb{P}(c_{i,n} = 0) = 1 - P_{i,n}^L. \quad (4)$$

Therefore, under the probabilistic LoS mode, the expected channel gain between the UAV and source GT i at time slot n is given by

$$h_{i,n} = c_{i,n}h_{i,n}^L + (1 - c_{i,n})h_{i,n}^N, \quad (5)$$

where $h_{i,n}^L = \beta_0 d_{i,n}^{-\alpha_L}$ and $h_{i,n}^N = \mu\beta_0 d_{i,n}^{-\alpha_N}$ indicate respectively the channel power gain in the LoS and NLoS cases, where β_0 represents the average channel power gain when the reference distance is 1 m in LoS state, $\mu < 1$ denotes the additional attenuation factor. In addition, α_L and α_N represent the average path loss exponents in LoS and NLoS states respectively, with $2 \leq \alpha_L \leq \alpha_N \leq 6$ [26]. Furthermore, $d_{i,n}$ denotes the Euclidean distance between the UAV and source GT i in time slot n , which is expressed as

$$d_{i,n} = \sqrt{\|\mathbf{q}_n - \mathbf{w}_{i,n}\|^2}. \quad (6)$$

In this paper, the communication between the UAV and GTs operates in a time division multiple access (TDMA) manners. Specifically, the UAV can only communicate with one GT during one time slot, and the scheduling is determined by binary scheduling variables $\alpha_{i,n}$ and $\alpha_{j,n}$ in time slot n , where $\alpha_{i,n} = 1$ or $\alpha_{j,n} = 1$ means that the UAV communicates with the i -th source GT or the j -th destination GT, respectively. The scheduling constraints can be denoted as

$$\sum_{i=1}^I \alpha_{i,n} + \sum_{j=1}^J \alpha_{j,n} \leq 1, \forall n \in \mathcal{N}, \alpha_{i,n} \in \{0, 1\}, \quad (7)$$

$$\alpha_{j,n} \in \{0, 1\}.$$

When the i -th source GT offloads the task to the UAV, the instantaneous achievable rate, denoted by $R_{i,n}$ in Mbits/second (Mbps), is expressed as

$$R_{i,n} = B_i \log_2 \left(1 + \frac{h_{i,n}P_i}{\sigma^2\Gamma} \right) = c_{i,n}R_{i,n}^L + (1 - c_{i,n})R_{i,n}^N, \quad (8)$$

where B_i is the transmission bandwidth of source GT i , P_i is the transmit power of source GT i , σ^2 represents the receiver noise power, $\Gamma > 1$ is the signal-to-noise ratio (SNR) gap between

the practical modulation-and-coding scheme and the theoretical Gaussian signaling. Moreover, $R_{i,n}^L = B \log_2(1 + \frac{\beta_0 P_i}{d_{i,n}^{\alpha_L} \sigma^2 \Gamma})$ and $R_{i,n}^N = B \log_2(1 + \frac{\mu \beta_0 P_i}{d_{i,n}^{\alpha_N} \sigma^2 \Gamma})$ represent the achievable rates from source GT i to the UAV in the LoS and NLoS states, respectively. It is worth noting that the channel power gain $h_{i,n}$ is affected by the randomness of small-scale fading and the occurrence of LoS and NLoS links. In our work, the effect of the small-scale fading is averaged out in each time slot by employing a channel code technique [26]. Then, by averaging the randomness source of occurring LoS and NLoS links, the expected achievable rate can be expressed as [26]

$$\mathbb{E}[R_{i,n}] = P_{i,n}^L R_{i,n}^L + P_{i,n}^N R_{i,n}^N. \quad (9)$$

Similarly, the instantaneous achievable rate from the UAV to the destination GT j can be denoted as

$$R_{j,n} = B \log_2 \left(1 + \frac{h_{j,n} P}{\sigma^2 \Gamma} \right) = c_{j,n} R_{j,n}^L + (1 - c_{j,n}) R_{j,n}^N, \quad (10)$$

where B denotes the UAV's transmission bandwidth, P expresses the transmission power of the UAV, $R_{j,n}^L = B \log_2(1 + \frac{\beta_0 P}{d_{j,n}^{\alpha_L} \sigma^2 \Gamma})$, $R_{j,n}^N = B \log_2(1 + \frac{\mu \beta_0 P}{d_{j,n}^{\alpha_N} \sigma^2 \Gamma})$ are the achievable rates from the UAV to the destination GT j in the LoS and NLoS states. The Euclidean distance $d_{j,n}$ between the UAV and destination GT j can be expressed as

$$d_{j,n} = \sqrt{\|\mathbf{q}_n - \mathbf{w}_{j,n}\|^2}. \quad (11)$$

Furthermore, we can express the expected achievable rate from the UAV to the destination GT j as

$$\mathbb{E}[R_{j,n}] = P_{j,n}^L R_{j,n}^L + P_{j,n}^N R_{j,n}^N, \quad (12)$$

where $P_{j,n}^L$ and $P_{j,n}^N$ denote the LoS probability and NLoS probability, respectively, between the UAV and the destination GT j .

C. UAV-Assisted Computing Model

After offloading a computation task from the source GTs, the UAV can immediately process it, as illustrated in Fig. 2. Specifically, the UAV processes the i -th GT's task after receiving the offloading from the $(i + 1)$ -th source GT. We assume that the task workloads are relatively low, resulting in computation times that are generally much shorter than offloading times. From Fig. 2, we can see that the pure processing time is the processing time of the last offloaded task. The size of the last source GT's workload is assumed W_i^* , so the total time slots to process the tasks can be represented as

$$N^C = \left\lceil \frac{C_{\dagger} W_i^*}{f_{\dagger} \sigma} \right\rceil, \quad (13)$$

where $\lceil \cdot \rceil$ rounds up to the nearest integer, $C_{\dagger}$ represents the number of rounds that the UAV's CPU turns when processing one sample, and $f_{\dagger}$ denotes the UAV's data processing ability, which mainly depends on its CPU's rotation speed in rounds per second. In particular, the given UAV computation capacity must be less than the UAV's maximum computational capacity, i.e.,

$f_{\dagger} \leq f_{\max}$, where $f_{\dagger}$ denotes the UAV's data processing ability, which mainly depends on its CPU's rotation speed in rounds per second, $f_{\max}$ is the maximum computational capacity.

D. Obstacle Avoidance Model

In the practical air-to-ground communication model, the UAV can obtain better CSI by flying within the feasible domain of unrestricted flight. This is because the UAV prefers to maintain a LoS channel for a higher transmission rate during its tasks. However, the distance between the UAV and GTs can be a limiting factor for offloading/transmission performance when the channel is in the LoS state, which suggests flying at a low altitude and closer to the GTs, i.e., flying at a low altitude. In other words, the UAV can perform its tasks below the height of the buildings. Thus, obstacle avoidance to 3D buildings should be taken into account for flight safety issues. In this work, we denote the total 3D spatial geographic area by Γ and the 3D spatial geographic area occupied by the building by Υ . Then, the feasible space for the UAV is expressed as

$$\Psi \triangleq \Gamma \setminus \Upsilon, \quad (14)$$

which represents the complement of Υ in the total 3D spatial geographic area Γ . Recall that GTs move on the ground in a 2D spatial geographic area $\mathcal{D}$, which is actually the two-dimensional plane that Ψ is mapped to the ground.

Remark 1: Although a 3D space UAV trajectory design increases the impact of large-scale fading in comparison to a 2D case, adjusting the UAV flying height is able to increase the LoS probability and make transmitting offloading tasks and decisions more efficient. Therefore, the 3D trajectory exploration scheme is capable of obtaining better performance than a 2D one.

III. PROBLEM FORMULATION

Our goal is to minimize the whole operation time via jointly optimizing the UAV's trajectory $\mathbf{q}_n$ together with the scheduling factors $\{\alpha_{i,n}\}, \{\alpha_{j,n}\}$ of offloading transmissions from source GTs and decision transmissions to destination GTs. Through the optimization of the UAV's trajectory and the scheduling of GT, the system's overall performance can be significantly enhanced³ [27], [28], [29], [30]. At the beginning of time slot n , we assume that the UAV knows the GTs' current locations from the locations' feedback by GTs, but the future GTs' locations are not known to the UAV. Based on this, we can establish the problem of whole operation time minimization as

$$\mathcal{P}1 : \min_{\{\mathbf{q}_n, \mathbf{v}_n, \alpha_{i,n}, \alpha_{j,n}\}} N\sigma \quad (15)$$

$$\text{s.t.} \quad \alpha_{i,n} \in \{0, 1\}, \forall i \in \mathcal{I}, n \in \mathcal{N}, \quad (15a)$$

$$\alpha_{j,n} \in \{0, 1\}, \forall j \in \mathcal{J}, n \in \mathcal{N}, \quad (15b)$$

³Upon analyzing (8) and (10) in our design, it is evident that optimizing the position of the UAV can significantly enhance the transmission rate, thereby reducing the whole operation time. Additionally, as depicted in Fig. 2, the data processing time $N^C \sigma$ is solely determined by the processing time of the last offloaded task. Therefore, by jointly optimizing the trajectory of the UAV and scheduling of GTs, the whole operation time can be reduced.

$$\sum_{i=1}^I \alpha_{i,n} + \sum_{j=1}^J \alpha_{j,n} \leq 1, \forall i \in \mathcal{I}, j \in \mathcal{J}, n \in \mathcal{N}, \quad (15c)$$

$$0 \leq z_n \leq H_{\max}, \forall n \in \mathcal{N}, \quad (15d)$$

$$\mathbf{q}_{n+1} = \mathbf{q}_n + \Delta_s \vec{\mathbf{v}}_n, \forall n \in \mathcal{N}, \quad (15e)$$

$$\Delta_s = V\sigma, \quad (15f)$$

$$\mathbf{q}_n \in \Psi, \forall n \in \mathcal{N}, \quad (15g)$$

$$\mathbf{w}_{i,n} \in \mathcal{D}, \mathbf{w}_{j,n} \in \mathcal{D}, \forall i \in \mathcal{I}, j \in \mathcal{J}, n \in \mathcal{N}, \quad (15h)$$

$$\sum_{n=1}^N \alpha_{i,n} R_{i,n} \geq \Xi_i, \forall i \in \mathcal{I}, n \in \mathcal{N}, \quad (15i)$$

$$\sum_{n=1}^N \alpha_{j,n} R_{j,n} \geq \Xi_j, \forall j \in \mathcal{J}, n \in \mathcal{N}, \quad (15j)$$

where (15a), (15b), and (15c) represent the scheduling constraint, and recall that N is the total number of time slots in which the UAV completed the task. Equation (15d) is the maximum flight altitude constraint of the UAV. Different from [26], [31], in this work the minimum flying height of the UAV no longer needs to be greater than the height of the tallest buildings, i.e., the UAV can perform its tasks between buildings. In addition, (15e) and (15f) are the flight displacement constraints of the UAV, where Δ_s represents the distance that the UAV moves per unit time slot, $\vec{\mathbf{v}}_n$ denotes the flight direction of the UAV in the slot n , and V is the speed of the UAV. Moreover, constraint (15g) means that the UAV needs to avoid obstacles in order to fly safely, and (15h) represents the feasible domain constraint of GTs moving on the ground. Furthermore, constraints (15i) and (15j) guarantee that the task of each source GT is fully offloaded and each decision information is successfully transmitted to the corresponding destination GT, respectively.

Despite the whole operation time T being discretized, problem $\mathcal{P}1$ is still a mixed-shaping non-convex problem. In addition, since the channel channel is affected by the locations of the UAV and the GTs, problem $\mathcal{P}1$ is significantly difficult to be analytically addressed, i.e., being challenging to be solved via conventional optimization methods [31], [32]. Moreover, the mobility of GTs also leads to dynamic changes in the system environment. In the next section, we propose a joint design based on a modified DRL approach to address the problem.

IV. PROPOSED φ_1 -STEP D3QN METHOD FOR UAV TRAJECTORY DESIGN

A. Modified DDQN Network Framework

RL is a potent technology that solves Markov decision processes (MDPs) through an iterative interaction between agents and the environment. MDPs comprise four essential components $\{\mathcal{S}, \mathcal{A}, \eta, r_n\}$, namely, the state space $\mathcal{S}$, the action space $\mathcal{A}$, the state transition probability η , and the reward function r . During time slot n , the agent observes the current state $s_n \in \mathcal{S}$, selects an action $a_n \in \mathcal{A}$ by following the policy η , and transitions to the next state $s_{n+1} \in \mathcal{S}$ while receiving a reward r_n . In particular,

the agent's objective is to maximize the cumulative discounted reward G_n , defined as

$$G_n = r_{n+1} + \gamma r_{n+2} + \gamma^2 r_{n+3} + \cdots = \sum_{k=0}^{\infty} \gamma^k r_{n+k+1}, \quad (16)$$

where $\gamma \in [0, 1]$ is the discount factor and its value will affect the final benefit obtained by the agent. However, the conventional Q-learning algorithm is a tabular learning method that entails storing learned action-state pairs in a table. This approach becomes impractical when the number of action-state pairs is exceedingly large since it requires substantial memory space to store the table, resulting in extremely high algorithmic complexity.

Deep reinforcement learning (DRL) combines the perception capabilities of DL with the decision-making abilities of RL. This integration enables direct control based on input, eliminates the need for tables to store learned data, and reduces algorithmic complexity. In DRL, the action-value function $Q_\pi(s, a) = \mathbb{E}_\pi[G_n | s_n = s, a_n = a]$ represents the expected return obtained by the agent when it takes action a_n in the current state s_n following policy $\pi(s)$. The action-value function $Q_*(s, a) = \max_\pi Q_\pi(s, a)$, optimized through continuous learning, satisfies the Bellman optimization equation

$$Q_*(s, a) = r(s, a) + \gamma \sum_{s^*} \eta(s^* | s, a) \max_{a^*} Q_*(s^*, a^*), \quad (17)$$

where s^* and a^* denote all future actions and states, respectively. In theory, if all transitions (s_n, a_n, r_n, s_{n+1}) are traversed, the optimal strategy $\pi_*(s_n)$ can be identified, and the optimal value $Q_*(s_n, a_n)$ can be obtained. The DDQN algorithm overcomes the limitations of table-based methods by utilizing DNNs to approximate the Q-value function. By employing two separate networks, it alleviates the disadvantage of difficulty in convergence that results from strong correlations of the neural network of the DQN algorithm [33]. The primary objective of the DDQN method is to minimize the loss function through training a neural network, thereby enhancing the accuracy of the neural network predictions. The loss function of DDQN is expressed as

$$L = (y - Q(s_n, a_n | \theta))^2, \quad (18)$$

where $y = r_{n+1} + \gamma Q(s_{n+1}, \arg \max_{a^*} Q(s_{n+1}, a^* | \theta^*) | \theta^*)$ is the real Q value, $Q(s_n, a_n | \theta)$ is the predicted value, θ and θ^* are the parameters related to the current Q network and the target Q network, respectively.

Nevertheless, the DDQN network only predicts state s_{n+1} according to the value $Q(s_n, a_n)$ of action a_n under the current state s_n . The prediction is inaccurate if the influence of the action a_n on the state s_n is small (there is not enough relationship between the two). In contrast to the DDQN network architecture, the dueling network architecture separates state and action. Even though the final result remains the same, the state is no longer completely dependent on the value of the action for judgment throughout the calculating process, and it can be judged individually. As a result, the proposed algorithm with the dueling networks becomes more stable and can obtain a superior performance. Besides, we also utilize multi-step bootstrapping to accelerate the convergences of the algorithm, making it converge

more quickly with a suitable multi-step φ_1 . The details of the above **modifications on the neural network** are presented as follows:

- *Dueling network*: The dueling network structure has two streams to evaluate the action value. One stream is the state value function, which is only utilized to predict the value of the state. Another stream is the state-dependent action advantage function, which is used to predict the quality of the action of the current state. Compared to traditional network architectures, it is able to learn which state is (or is not) valid without learning the quality of the action for each state [34]. Therefore, it is more stable and can obtain a superior performance. The Q-function of the dueling network architecture can be expressed as

$$Q(s, a|\theta, \phi, \beta) = V(s|\theta, \beta) + A(s, a|\theta, \phi) - \frac{1}{|A|} \sum_{a^*} A(s, a^*|\theta, \phi), \quad (19)$$

where $V(s|\theta, \beta)$ denotes the state value and $A(s, a|\theta, \phi)$ is the action advantage value. In addition, β and ϕ are the parameters of the state value stream and the state-dependent action advantage stream and $|A|$ represents the size of action space.

- *Multi-step bootstrapping*: In comparison to Monte Carlo (MC) and temporal difference (TD), the multi-step bootstrapping approach considers the return of the future φ_1 steps,⁴ and selecting an appropriate φ_1 enables the method to converge faster and better [36]. The truncated φ_1 -step return is defined as

$$R_{n:n+\varphi_1} = \sum_{k=0}^{\varphi_1-1} \gamma^k r_{n+k+1}. \quad (20)$$

Based on the above improvements, the target Q value can be expressed as

$$L = \left(R_{n:n+\varphi_1} + \gamma^{\varphi_1} Q^*(s_{n+\varphi_1}, \arg \max_{a^*} Q(s_{n+\varphi_1}, a^*|\theta)|\theta^*) - Q(s_n, a_n|\theta) \right)^2, \quad (21)$$

where Q^* is the Q function value of the target network.

Remark 2: In RL, TD and MC are two basic learning approaches. The TD method only utilizes the current state information to update the Q value, which makes the final prediction result inaccurate. On the contrary, MC adopts all the future state information to update the Q value. Once the MC technique is adopted, the converging speed of the method is highly influenced. In particular, when there are too many future states, the speed is more likely incredibly slow, which leads to a negative impact on the final converged results. Note that the multi-step method utilizes the next φ_1 -steps to update the Q value. Hence,

⁴The φ_1 -step refers to the future movement steps of the UAV [35]. Specifically, if the UAV receives a reward of r_n at time slot n and rewards of $r_{n+1}, r_{n+2}, \dots$, and $r_{n+\varphi_1}$ in the subsequent φ_1 time slots, the truncated return of slot n can be expressed as Formula (20).

choosing an appropriate φ_1 is expected to provide significant performance enhancements on the method.

B. φ_1 -Step D3QN-Based DRL Solution to $\mathcal{P}1$

In the DRL model, the UAV acts as an agent, which selects an action to interact with the environment according to the current state and obtains a reward, and finally moves to the next state. When the UAV decides on an action, the ε -greedy strategy is utilized to strike a balance between exploration and exploitation so that the UAV can swiftly determine the best action. The ε -greedy strategy can be expressed as

$$a = \begin{cases} \arg \max Q_{\cdot a} & \vartheta > \kappa, \\ \text{randomly choose an action,} & \vartheta \leq \kappa, \end{cases} \quad (22)$$

where $\vartheta \in [0, 1]$ is a randomly generated number, and $\kappa \in [0, 1]$ is exploring probability that mainly balances exploration and exploitation. In detail, before the UAV selects an action, a decimal ϑ from zero to one is randomly generated. When ϑ is greater than κ , the UAV will select the optimal action according to the optimized Q value. Conversely, the UAV randomly chooses an action from the action space $\mathcal{A}$ when ϑ is less than κ . In this paper, the state, action, reward function and the state transition model in detail follows:

1) *State s_n* : We consider the state s_n as the three-dimensional coordinates $[x_n, y_n, z_n]$ of the UAV during flight, where the state space S contains all the states in the feasible region.

2) *Action a_n* : In this paper, the action a_n of the UAV corresponds to the flight direction of the UAV, which has six flight directions $[a_1, a_2, a_3, a_4, a_5, a_6]$, where $a_1 = [0, 1, 0]$ means that the UAV moves to the forward, $a_2 = [0, -1, 0]$ represents the UAV moves to the behind, $a_3 = [1, 0, 0]$ denotes the UAV moves to the left, $a_4 = [-1, 0, 0]$ means the UAV moves to the right, $a_5 = [0, 0, 1]$ means the UAV is flying up, $a_6 = [0, 0, -1]$ denotes the UAV is flying down.

Lemma 1: In problem $\mathcal{P}1$, if the total size of the data to be offloaded and the size of the transmission decision information $Q = \sum_{i=1}^I \Xi_i + \sum_{j=1}^J \Xi_j$ is a constant, and the whole operation time is discrete at equal intervals, then Problem $\mathcal{P}1$ is equivalent to $\mathcal{P}2$, i.e. $\min N\sigma \Leftrightarrow \max R_{k,n}, \forall k \in \mathcal{I} \cup \mathcal{J}$.

$$\mathcal{P}2 : \max_{\{\mathbf{q}_n, \bar{\mathbf{v}}_n, \alpha_{i,n}, \alpha_{j,n}\}} R_{k,n} \quad (23)$$

$$\text{s.t. (15a) - (15j)}, \quad (23a)$$

Proof: To prove that Lemma 1 holds, we will use a proof by contradiction. Specifically, we will show that the inverse proposition of Lemma 1 is false. It's worth noting that the pure processing time of the first stage is equivalent to the processing time of the last offloaded task. This time is not affected by the transmission rate and is only dependent on the GTs' scheduling. As a result, the data processing that occurs in the first stage has no bearing on the proof of Lemma 1. Then we write down the inverse proposition of Lemma 1, recall that if the total size of the data to be offloaded and the size of the transmission decision information $Q = \sum_{i=1}^I \Xi_i + \sum_{j=1}^J \Xi_j$ is a constant, and the whole operation time is discrete at equal intervals, then

$\min N\sigma \not\Leftarrow \max R_{k,n}$ can be deduced. To prove this, we first demonstrate that $\min N\sigma \not\Leftarrow \max R_{k,n}$ is false. Consider the case where there is only one GT, $Q = Y$, and $0 < R_{k,n}\sigma \leq Y$, where Y is a constant. The constraint $\sum_{n=1}^N R_{k,n}\sigma \geq Y$ implies that the optimal solution for problem $\mathcal{P}1$ is achieved when $N = 1$, and only if $R_{k,n} = Y/\sigma = \max R_{k,n}$. Thus, we conclude that $\min N\sigma \not\Leftarrow \max R_{k,n}$ is false. Similarly, we can show that $\min N\sigma \not\Leftarrow \max R_{k,n}$ is also false. Therefore, the inverse proposition of Lemma 1 does not hold, and the proof is completed. $\square$

3) *Reward r_n* : To ensure obstacle avoidance constraints for the UAV in problem $\mathcal{P}2$ with respect to 3D buildings and expedite task completion, we have designed the reward function reasonably. According to Lemma 1, the transmission rate $R_{k,n}$ of each time slot in the data offloading from source GTs and decision transmission to destination GTs is the key factor impacting the whole operation time. In addition, considering that the UAV also needs to avoid obstacles when completing its tasks in the shortest time, we set the reward function r_n at time slot n as follows:

$$r_n = \begin{cases} \Lambda, & \text{if the UAV out of bounds or hit buildings;} \\ \phi * \mu * N^C * \sigma + \lambda * R_{k,n}, & k \in I \cup J, \text{ else,} \end{cases} \quad (24)$$

where ϕ is a binary variable to judge whether the first stage is over, i.e., $\phi = 1$ represents the first stage is over, otherwise, $\phi = 0$. In addition, μ is a negative value in order to reduce data processing time, Λ is a negative value for the penalty when the UAV goes out of bounds or hits buildings, and λ is the throughput coefficient. Therefore, by designing reasonable values for μ , Λ , and λ , the UAV can satisfy constraints and complete tasks quickly.

4) *State Transition Model*: The state transition model represents the probability that when the UAV is in state s_n , it chooses action a_n according to policy π , and then moves to the next state s_{n+1} . The state transition model is deterministically governed by (15e).

Pseudo-codes for UAV trajectory design and communication scheduling are provided by Algorithms 1 and 2. In the preparation phase of the Algorithm 1, we first initialize various parameters, such as the replay buffer, the parameters θ and θ^* of the main network and target network, and so on. The initial location $\mathbf{q}_0$ and the GTs' initial location $\mathbf{w}_{i,0}, \forall i \in \mathcal{I}$ and $\mathbf{w}_{j,0}, \forall j \in \mathcal{J}$ are randomly generated (line1). Then there is the process by which the UAV interacts with the environment. At the beginning of each episode, the UAV executes Algorithm 2 based on the current position and determines the GTs scheduled for the current slot. The source GT then offloads the data to the UAV for processing or the UAV transmits the decision data to the destination GT with ID_k .

After source GTs complete the data offloading or the UAV transmits the decision information to the destination GT of the current time slot, the UAV chooses an action utilizing ε -greedy strategy from the action space. In the early stage of UAV training, the UAV is given a significant exploration probability, which is gradually reduced, and then increased the probability of exploitation, in order to the UAV thoroughly explore and find the optimal strategy quickly (line 2–8). The initial exploration

Algorithm 1: Proposed D3QN Method-Based Approach for Jointly Optimizing UAV Trajectory Design and GTs Scheduling.

```

1: Initialize : the replay buffer with space  $C$ , the
   parameters of dueling DQN main network and target
   network  $\theta, \theta^*$ , the initial exploration probability  $\varepsilon$ , the
   exploration decay rate  $\alpha_\varepsilon$ , the initial location of the UAV
    $\mathbf{q}_0$ , and GTs' initial location  $\mathbf{w}_{i,0}, \forall i \in \mathcal{I}$  and
    $\mathbf{w}_{j,0}, \forall j \in \mathcal{J}$ .
2: for episode = 1, 2, ...,  $L$  do
3:   Set: the sliding window queue  $E$  with space  $\wp_1$  and
       $n \leftarrow 0$ .
4:   while the task not completed do
5:     The UAV execute Algorithm 2 to obtain  $\mathbf{o}_k^*$ .
6:     The UAV receives the offloading data or transmits
       the decision information from the GT with  $ID_k$ .
7:     Select action  $a_n$  from action space  $\mathcal{A}$  based on the
        $\varepsilon$ -greedy policy.
8:     Execute action  $a_n$  to obtain reward  $r_n$  and move
       to the next state  $s_{n+1}$ .
9:      $\tau \leftarrow n - \wp_1 + 1$ ;
10:    if  $\tau \geq 0$  then
11:      Calculate the  $\wp_1$ -step reward  $R_{\tau:\min(\tau+\wp_1, \wp_1)}$ 
        based on (20);
12:      Stored the transition
         $(s_\tau, a_\tau, R_{\tau:\min(\tau+\wp_1, \wp_1)}, s_{\min(\tau+\wp_1, \wp_1)})$  to  $C$ .
13:    end if
14:    Randomly select mini-batch of transition
       $(s_k, a_k, R_{k:k+\wp_1}, s_{k+\wp_1})$  from  $C$ ;
15:    Utilize the gradient descent method to update the
      evaluation network parameter  $\theta$  based on (21);
16:    Update  $n \leftarrow n + 1$ ;
17:    Every  $M$  steps reset the target network parameters
       $\theta^* \leftarrow \theta$ .
18:  end while
19:   $\varepsilon \leftarrow \varepsilon \alpha_\varepsilon$ ;
20: end for

```

rate is ε , which decays at a rate of α_ε and is not less than 0.01. The UAV performs an action and calculates the throughput of data collected or sent in this time slot, then moves to the next state and gets a reward based on (24). The truncated return of $\wp_1$ -step is computed and $(s_\tau, a_\tau, R_{\tau:\min(\tau+\wp_1, \wp_1)}, s_{\min(\tau+\wp_1, \wp_1)})$ is stored in replay memory C . Then randomly sample a mini-batch transition from replay memory C , and utilize gradient descent to update the evaluation network parameter θ . The parameters θ of the evaluation network are copied to the target network every M step period. Finally, the next episode begins if the UAV successfully completes its task; otherwise, lines 5–17 repeat (lines 9–17).

Fig. 3 presents the framework of the proposed method. Specifically, at any given time slot n , the UAV obtains the predicted Q-value $Q(s_n, a_n; \theta)$ from the current dueling network based on its current state s_n and action a_n . In particular, θ is obtained from training the neural network utilizing the historical position

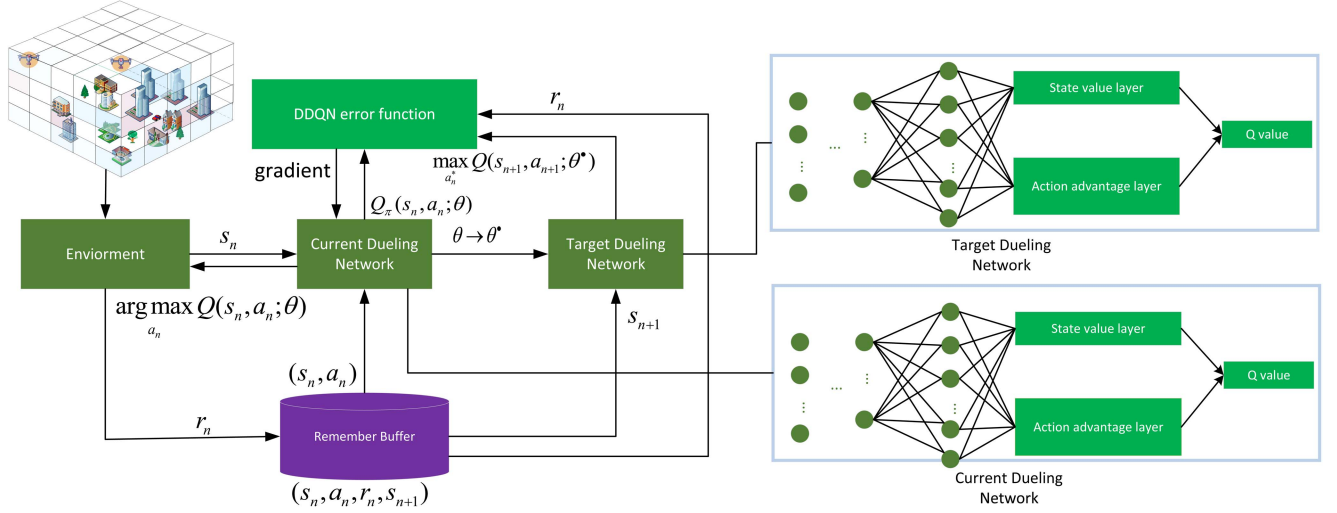


Fig. 3. Framework of the proposed method for UAV 3D trajectory design.

data of the UAV and the GTs. Subsequently, all the possible next state s_{n+1} is fed into the target dueling network, which yields different Q-values based on different actions and the UAV selects the a_n^* that maximizes the Q-value $Q(s_{n+1}, a^* | \theta)$. Next, the gradient descent method is employed to minimize the loss function L , i.e. Equation (21), with the objective of minimizing the disparity between predicted values and actual values. It can be observed that the loss function L is influenced by the reward function r_n , i.e., (24), which is mainly determined by whether the UAV is flying safely, the computation time N^C , and the transmission rate $R_{k,n}$ associated with the UAV's location and the GTs location. Consequently, in order to complete the task faster and obtain more rewards, the UAV will adjust its location based on the GTs' location to maximize the transmission rate $R_{k,n}$ and also avoid collisions.

Remark 3: The size of the mini-batch determines the utilization of samples and the convergence performance of the proposed method during the learning process. Because the data stored in the replay buffer cannot be effectively utilized when the sample size is improperly chosen, the proposed method converges too slowly with a poor outcome. Therefore, it is crucial to choose a suitable sample size when the UAV interacts with the environment.

C. Convergence and Complexity Discussions

1) *Convergence of the Proposed Method:* To prove the convergence performance of the proposed method, we first need to prove the convergence performance of the double Q-learning method, and then we require to analyze the effect of multi-step bootstrapping and the dueling networks on the method's convergence. In the theorem 1 of [38], it has been proved that the double Q-learning method can converge to the optimal value under certain conditions, such as $0 \leq a_n \leq 1$, $\sum_n a_n = \infty$, $\sum_n a_n^2 < \infty$. In addition, the convergence of the method also depends on the design of the hyperparameters. The exploration rate of the environment by the UAV in our proposed method decreases with the number of iterations, and we also utilize

Algorithm 2: GTs Scheduling Optimization

- 1: **Initialize :** The ID of GTs $ID_k, \forall k \in \mathcal{I} \cup \mathcal{J}$, the GTs' transmitting rate $R_{k,n}, \forall k \in \mathcal{I} \cup \mathcal{J}$, the remaining data size of the tasks $D_k^*, \forall k \in \mathcal{I} \cup \mathcal{J}$.
- 2: Store $R_{k,n}, D_k^*$, and ID_k in matrix $P = [\mathbf{o}_1, \dots, \mathbf{o}_K] = [[R_{1,n}, C_1^*, ID_1], \dots, [R_{K,n}, C_K^*, ID_K]]$.
- 3: **for** $i = 1, 2, \dots, K$ **do**
- 4: **if** $C_k^* \leq 0$ **then**
- 5: Remove $[R_{k,n}, D_k^*, ID_k]$ from P .
- 6: **end if**
- 7: **end for**
- 8: Sort $\mathbf{o}_k$ from largest to smallest with $R_{k,n}$ as the standard, i.e. $P^* = [o_k^*, \dots, o_{k+m}^*]$.
- 9: **if** the first stage is completed **then**
- 10: **Output:** The largest $\mathbf{o}_k^*, k \in \mathcal{J}$.
- 11: **end if**
- 12: **if** the first stage is not completed **then**
- 13: Obtain the largest $\mathbf{o}_k^*$.
- 14: **while** $k \notin \mathcal{I}$ **do**
- 15: $k = k + 1$
- 16: **end while**
- 17: **Output:** The largest $\mathbf{o}_k^*, k \in \mathcal{I}$.
- 18: **end if**

gradient descent to update the model parameters θ . The neural network's convergence performance is guaranteed after a pre-determined number of iterations because the UAV will stop exploring the environment and its model parameters won't change θ . On the other hand, in the process of improving the network architecture, multi-step bootstrapping and dueling networks will only change the method's stability and convergence speed, but not change the convergence performance of the proposed method [37]. As a result, the convergence performance of D3QN method is guaranteed. Besides, we also provide the method's convergence and robustness through simulation results in this paper.

TABLE I
SIMULATION PARAMETERS

Para	Description	Value	Para	Description	Value
P	Transmission power of the UAV	0.1(W) [32]	P_k	Transmission power GTs	0.1(W) [32]
B	Total channel bandwidth	10(MHz) [32]	σ^2	Noise power	-109(dBm) [26]
β_0	Average channel power gain	-60(dB) [26]	Γ	The SNR gap	8.2(dB) [26]
α_N	Average path loss exponents(NLoS)	3.5 [26]	α_L	Average path loss exponents(LoS)	2.5 [26]
T	Update frequency for target network	10	σ	One time slot	0.5(s) [37]
μ	Additional attenuation factor	-20(dB) [26]	C	Replay buffer	3×10^5
ε	Initial exploration probability	0.8	α_ε	Exploration decay rate	0.9954
L	The maximum episodes	1000	$\wp_1$	The sliding window space	64

2) *Complexity of the Proposed Method:* The computational complexity of the proposed method mainly includes the learning process and exploration [24]. The fully connected layer of the neural network primarily influences the former, whereas the UAV's exploration space and step size mainly affect the latter [24]. The neural network in our design has five fully connected layers. First, the complexity of the fully connected layer neural network model is $\mathcal{O}(\sum_{l=1}^L f_l f_{l-1})$, where f_l is the number of neural units of the fully connected layer l and L is the number of fully connected layers. Secondly, the complexity of the UAV in the exploration and learning process can be expressed as $\mathcal{O}(|S||A|)$ [24], where $|S|$ represents the total state of the UAV in the exploration process with step size Δ_D , and $|A|$ denotes the size of the UAV's action space.

V. SIMULATION RESULTS

A. Simulation Settings

In the simulation, we consider a regularly distributed dense 3D urban environment with a size 1000×1000 m, where the maximum height of the building is 50 m. The number of source GTs is set to $I = 3$, and the number of destination GTs is set to $J = 3$. In addition, the initial locations of the UAV and the GTs are randomly generated on the ground. The parameter B_1, B_2, B_3 and B_4 in formula (3) are set to $B_1 = -0.4568$, $B_2 = 0.0470$, $B_3 = -0.63$ and $B_4 = 1.63$, respectively [26]. The movement speed of GTs is 5 m/s and the speed of the UAV is $V = 25$ m/s. Besides, the maximum flying height of the UAV is set to $H_{\max} = 100$ m. The data size that the source GTs S_1, S_2 , and S_3 need to offload are $\Xi_{S_1} = 100$ Mbit, $\Xi_{S_2} = 300$ Mbit, and $\Xi_{S_3} = 200$ Mbit, respectively. Besides, the workloads of the task that need to be processed are $W_1 = 5$ Mcycles, $W_2 = 20$ Mcycles, and $W_3 = 10$ Mcycles. The CPU frequency of the UAV is 4 GHz. The data sizes of the decision information that the UAV needs to transmit to the destination GTs are set to $\Xi_{D_1} = \Xi_{D_2} = \Xi_{D_3} = 600$ Mbit. Moreover, the channel parameters and the Algorithm parameters are shown in Table I.

In addition, the reward's parameters μ, r_b and the throughput coefficient λ are $\mu = -20$, $r_b = -20$, and $\lambda = 10$, respectively. The evaluation network and the target network have the same network structure. They all have five hidden layers, and the number of neurons in the first four hidden layers are 512, 256, 128, and 128. The last hidden layer is the dueling layer with 7 neurons,

with one neuron corresponding to the estimation state-value and the other 6 neurons corresponding to the action advantage for 6 actions. The activation function of the hidden layer is the ReLu function and the optimizer is Adam. The operating environment is Python 3.7 with TensorFlow 2.0.0 and Keras 2.3.1.

B. Validation and Evaluation

In this section, we first investigate the convergence performance of the algorithm, and then select appropriate hyperparameters for the proposed algorithm. Furthermore, we evaluate the performance of the proposed algorithm in terms of the whole operation time (the objective of our design), 3D trajectory results, GT scheduling results and the robustness of the algorithm. It should be mentioned that in the following simulations, two scenarios with different random moving behaviors of GTs are considered. At any time slot n , we assume that the UAV can acquire historical position information of GTs through pre-existing feedback from the GTs or sensors deployed on the ground [39], [40], but the future locations are currently unknown to the UAV. In particular, our proposed algorithm is independent of any specific GTs movement model. The two models presented in this section are only utilized to demonstrate the efficacy of the proposed algorithm, and they are as follows:

- Scenario with GTs randomly moving in a small area (SA): GTs move in an area randomly and frequently change their directions and speeds. For example, these GTs are taking pictures or searching/detecting objectives in a small region, etc.
- Scenario with GTs moving along a relatively long path (LP): GTs move on a ground path and less frequently change their directions and speeds. For instance, these GTs are delivering cargo from their initial location to another location during the process of receiving or transmitting data.

1) *Convergence Validation:* We start with Fig. 4(a) and (b) to investigate the convergence performance of the proposed $\wp_1$ -step D3QN method in the SA scenario, while the impacts of the moving average return and the whole operation time are studied, respectively. In particular, we obtain the results of moving average return by calculating an average reward per every 100 episodes in 10,00 episodes. The results of different curves are conducted under different and independent experiments, while the initial locations of the UAV and GTs are randomly generated

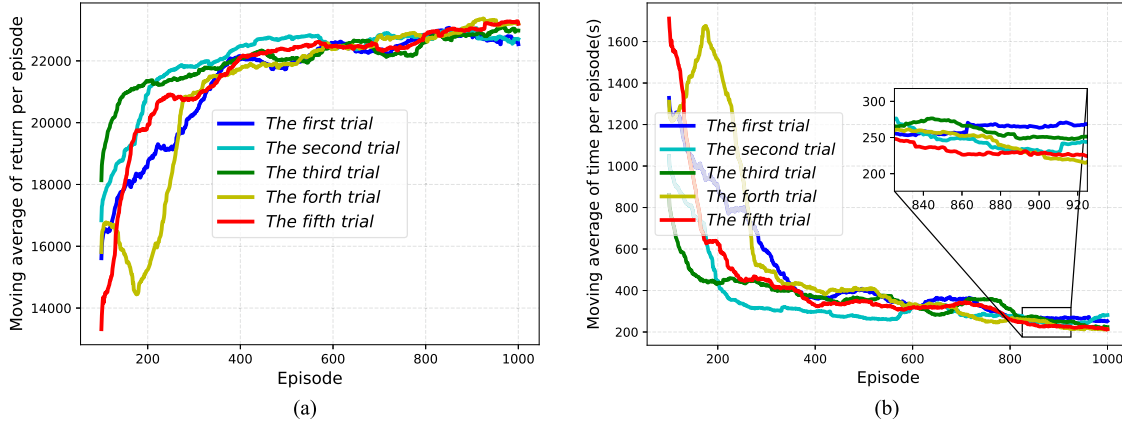


Fig. 4. Convergence investigation. (a) Reward of scenario SA. (b) The whole operation time of scenario SA.

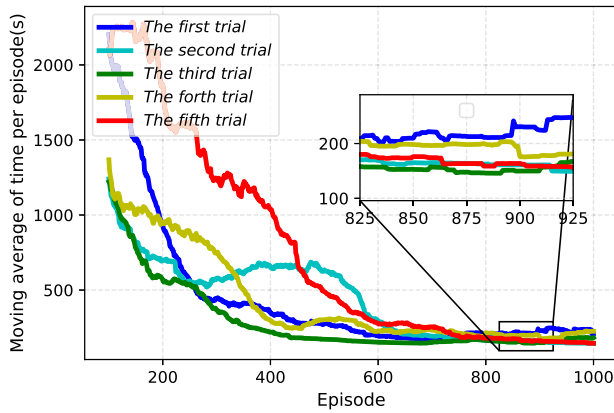


Fig. 5. Whole operation time of scenario LP.

in these experiments. One can observe from Fig. 4(a) and (b) that for different initial positions of the UAV and GTs, the moving average return and the whole operation time of these simulation experiments eventually converge to the approximate value, which confirms the convergence performance of the proposed algorithm. Additionally, it can also be seen from Fig. 4(b) that the whole operation time initially gradually decreases from a higher value before converging to a stable value. The explanation is that in the process of interacting with the environment, the UAV is more likely not able to effectively learn a sufficient efficient action strategy due to a lack of experience in the early stage. As the experience grows, the UAV could gradually improve the efficiency of the action selection strategy, which results in gradually decreasing the whole operation time. In particular, we also can see the final convergence values of the five curves in Figs. 4 and 5 are slightly different. This primarily stems from the different initial locations of GDs, network parameters, and the time-varying exploration behavior of these five training curves.

Next, Fig. 5 illustrates the convergence performance of the whole operation time for UAV 3D trajectory design in the LP scenario. Each curve in the figure also represents an independent experiment, and in all experiments, the UAV and GTs have different initial positions. In comparison to scenario SA, the whole operation time of scenario LP converges faster, e.g., scenario SA converges at 800 episodes as shown in Fig. 4(b), while scenario

LP converges at 700 episodes as shown in Fig. 5. Moreover, the whole operation time of scenario LP is less than scenario SA. This is primarily due to the fact that all GTs gradually move around the center cluster of buildings. To adapt to the movement of GTs, the UAV only needs to fly around the middle cluster of buildings, which will be shown in Fig. 10. In comparison to scenario SA, the UAV can adjust its own trajectory faster to adapt to the movement of GTs, and the distance that the UAV flies is shorter, therefore the whole operation time is shorter and the convergence is faster.

2) *Algorithm Hyperparameter Selection:* In this subsection, we compare different hyperparameters of the algorithm and select the appropriate one in scenario SA. Fig. 6(a) plots the relationship between the whole operation time and learning rate. The initial locations of the UAV and GTs are identical in these three curves. Fig. 6(a) demonstrates that the learning rate of 0.0001 converges faster than the learning rate of 0.001 and 0.00001, and is more oscillation-stable. This is due to the fact that when the DQN network is trained, it is simple to miss the optimal value when the learning rate is too high, while when the learning rate is too low, it is easy to fall into the local optimal solution; hence, when the learning rates are 0.001 and 0.00001, the oscillation is more severe and the convergence effect will be worse. For instance, during a training regimen spanning 620 episodes with a learning rate of 0.0001, a notable anomaly occurs in the training curve. This phenomenon is attributed to the diminutive learning rate, which prompts the UAV to become ensnared in a local solution while undergoing training. Therefore, in our simulation, the learning rate is selected as 0.0001, which can better guarantee that the performance of the algorithm converges to the optimal value.

The results in Fig. 6(b) depict the relationship between the whole operation time and the size of φ_1 . In all three curves, the initial locations of the UAV and GTs are the same. Fig. 6(b) demonstrates that different φ_1 have various effects on the algorithm's convergence results. In comparison to $\varphi_1 = 16$ and $\varphi_1 = 256$, the whole operation time of $\varphi_1 = 64$ not only has a lower final convergence value, but also less oscillation in the overall process and is more stable. This is because, while training the DQN network, a suitable φ_1 can discover the optimal Q value. When φ_1 is too small, only fewer states are utilized to

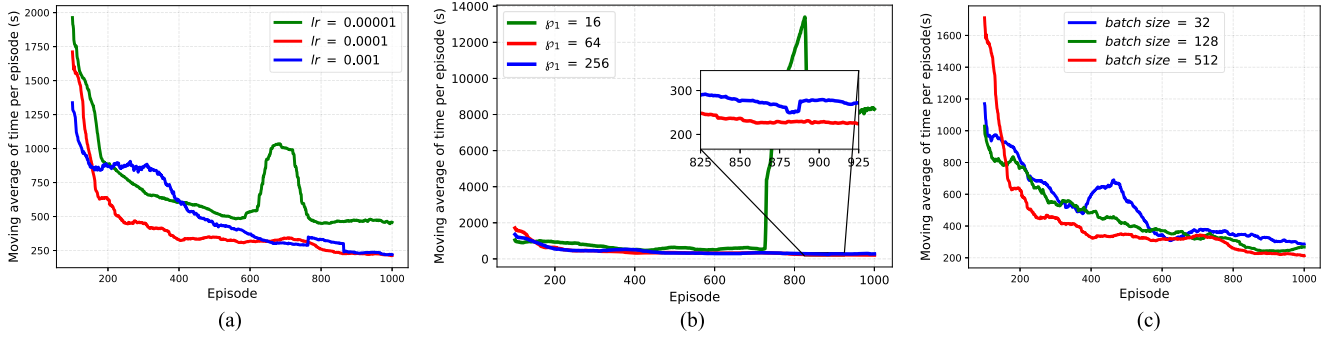


Fig. 6. Algorithm hyperparameter selection. (a) Learning rate vs. the whole operation time. (b) φ_1 vs. the whole operation time. (c) Batch size vs. the whole operation time.

update the current Q value, resulting in inaccurate predictions, while when φ_1 is too large, the algorithm will converge slowly and the ultimate convergence value is poor. This phenomenon is also confirmed by the insights provided in Remark 2. We choose $\varphi_1 = 64$ in our simulation trials to ensure the performance of our proposed algorithm.

Fig. 6(c) shows the relationship between the whole operation time and the size of mini-batch samples. In all three curves, the UAV and the GTs have the same initial location. It can be seen that the curve converges the fastest and the whole operation time is the shortest when the batch size is 512. This is due to the fact that a larger batch size can fully utilize the sample data stored in the replay buffer and makes it easier for the UAV to find the optimal action selection strategy when interacting with the environment. Compared with the batch size of 512, when the batch size is 128 or 32, it oscillates more strongly in the process of starting training. Because the sample is not fully utilized when the UAV initially interacts with the environment, and the UAV's action selection strategy is sub-optimal, the whole operation time of the UAV fluctuates. This phenomenon is also confirmed by insights in Remark 3. We selected the batch size as 512 for our simulation studies in order to fully utilize the samples and ensure the performance of our proposed algorithm.

3) *Performance Evaluation of the Proposed Method:* We move on to evaluate the performance of the proposed method. In particular, we conduct a comparative analysis between our design's 3D trajectory and the 2D trajectory, which requires the UAV to maintain a fixed altitude. Through this analysis, we demonstrate the advantages of an unrestricted feasible domain for UAV flight and explore the differences in trajectory behavior under two distinct GTs movement scenarios.

First, in Fig. 7 we provide the comparison between our design with the 2D trajectory one, while the scenario SA is considered. Obviously, the whole operation time under our design with a 3D UAV trajectory is shorter than the 2D cases, which indicates that the adjustable height of the UAV helps to achieve an efficient trajectory. This is primarily due to the increased flexibility that 3D trajectory design offers in balancing angle and transmission distance. The results match well with the discussions in Remark 1. On the other hand, our design with a 3D trajectory exploration converges at about 800 episode, which is slower than that the 2D trajectory design. This is due to the fact that the UAV's exploration space of 3D trajectory design

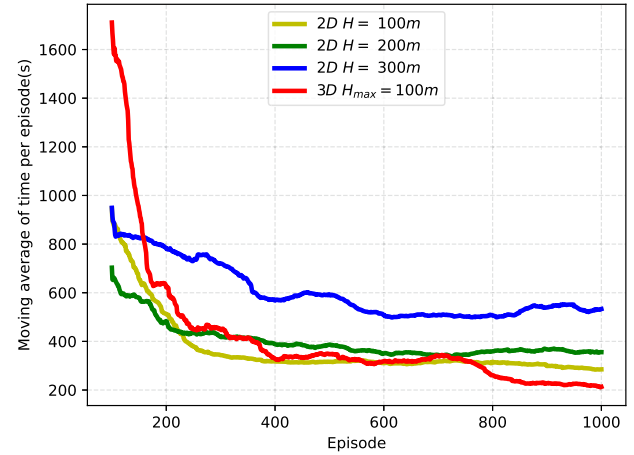


Fig. 7. Whole operation time in 2D exploration and 3D exploration.

is larger than that of 2D case, i.e., more episodes are required in the exploration to achieve an efficient solution. Besides, we learn from the figure that the whole operation time under a 2D trajectory design is generally increasing the fixed UAV altitude. The reason is that the communication connections between the UAV and GTs have a high LoS probability when the UAV flies at a sufficient high altitude, i.e., the transmission rate is mainly influenced by the transmission distance.

We study the impact of the allowed flying altitude ranges of UAV on the whole operation time. The results are provided in Fig. 8 where three different ranges are considered: a) range [0–50] m indicating that the UAV can fly between the ground and the tallest building, b) range [50–100] m indicating that the minimum flying height of the UAV is greater than the maximum height of the building, and c) range [0–100] m representing the case that the UAV can fly between buildings and above them. Clearly, obstacle avoidance to 3D buildings should be considered in the first two cases. In addition, the initial locations of the UAV and GTs for the three cases are the same. From Fig. 8, we can observe that for the three different flight domains, the proposed method converges rapidly, indicating that our improved algorithm possesses strong adaptability. In addition, we learn from the results that given the same initial conditions, case c), i.e., the one that enjoys the highest freedom in terms of flying altitude, achieves the shortest whole operation time, followed by case a), while case b) is the worst one in terms of the whole operation

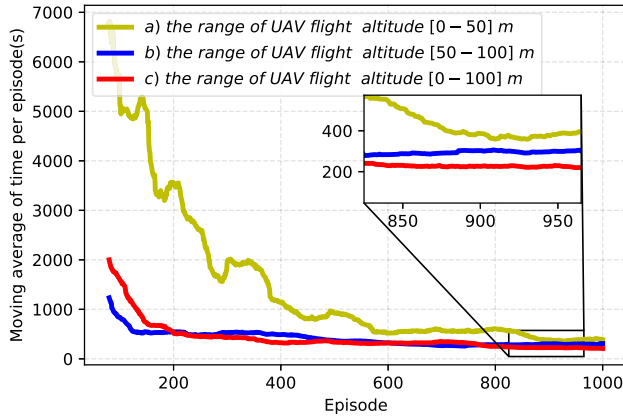


Fig. 8. Whole operation time vs. the number of training episodes with different flight altitude ranges.

time. The reason are as follows. Due to the buildings' obstruction when carrying out the mission in case a), it is challenging for the UAV to establish a reliable communication connection with GTs. In case of b) where the flying height could be high, although the links between the UAV and GTs are more likely to be LoS channels, the transmission distances between the UAV and GTs are relatively long, i.e., being longer than the minimal allowed UAV height of 50 m. The results demonstrate again the benefits of enjoying a flexible flying altitude range. In addition, it can be observed from the figure that in comparison with case b) and case c), the convergence performance of case a) is low in terms of converge speed and stability (the curve of range [0–50] m oscillates violently). This is due to the fact that the heights of buildings are set to 50 m, which indicates that in the case a) (the UAV flying altitude range is [0–50] m) the UAV always flies among buildings, and the method operates suffering from the obstacle avoidance constraint all the time. This is also the reason why the performance of case c) is relatively lower than case b) without sufficient episodes.

Next, we plot the designed UAV 3D trajectories in Figs. 9 and 10 under the SA scenario and the LP scenario, respectively. The source GTs are represented by S_1 , S_2 , and S_3 , and destination GTs are indicated by D_1 , D_2 , and D_3 , as shown in Figs. 9 and 10. In both figures, each red 3D curve represents a UAV's 3D trajectory, while other curves indicate GTs' trajectories. From the figures, it can be seen that our method successfully ensures the UAV avoids 3D buildings during the mission under different GTs mobile models. This indicates that the improved algorithm can adapt to complex 3D obstacle avoidance environments. The figures also illustrate that the UAV first raise altitude after taking off, and then flies close to the served GT. For example, from Fig. 9 we can see that the UAV raises its height when the source GT S_1 offloads tasks to the UAV, then the UAV approaches GT S_3 and GT S_2 gradually in the first operation stage. The explanation of this is as follows. While minimizing the whole operation time, a trade-off between the angle and the distance from its serving GT should be considered. On the one hand, the UAV improves the LoS probability of servicing GTs by increasing its own flight altitude. On the other hand, the communication transmission rate is increased by the proximity of the UAV to the GT it serves. As shown in Fig. 10, we also can see that the

GTs travel gradually around the middle cluster of buildings, and the GTs have a greater movement range than in scenario SA. Different from the scenario SA, the UAV primarily flies over the center cluster of buildings after raising its height in the scenario of LP. This indicates that the UAV follows the moving GTs to fly over the middle cluster of buildings during the mission. The results in both figures actually confirm the robustness of the proposed design, i.e., our method is able to adapt to the different GTs moving behavior models.

The corresponding scheduling results of Figs. 9 and 10 are provided in Fig. 11(a) and (b), respectively. In the two figures, the scheduling factor having the value of 1 indicates that the GT is scheduled, otherwise, it is not scheduled. Combining Fig. 9 with Fig. 11(a) or combining Fig. 10 with Fig. 11(b), we can see that the UAV usually first schedules the closest GTs, followed by the GTs farther away. In this process, the UAV gradually closes to the scheduled GTs. For instance, from Fig. 9 with Fig. 11(a), we can see that in the first stage of the system operation, the source GTs S_1 , S_3 , and S_2 offload their tasks to the UAV successively. During the process of data offloading, the UAV also gradually flies to the source GTs S_1 , S_3 , and S_2 . This is due to the fact that the UAV will prioritize scheduling the GT with a higher transmission rate and gradually approach the scheduled GT to improve the quality channel to decrease the whole operation time. In addition, we can see from Fig. 11 that the UAV does not always choose a specific destination GT for decision transmission in the decision transmission stage. The reason for this is as follows. Recall that when the UAV adjusts its position to adapt to the movement of the GT, it will prioritize scheduling the destination GT with a higher transmission rate based on its current location. Once the UAV moves to the next location and the destination GT of the maximum communication rate changes, the UAV will switch scheduled destination GT, i.e., the UAV schedules D_3 from 43 s to 62 s and then schedules D_2 from 63 s to 88 s as shown in Fig. 11(a). Moreover, it is interesting that the scheduling order results for decision transmissions are different in the two scenarios, i.e., the UAV schedules D_3 first in scenario SA, while the UAV schedules D_2 in scenario LP. This is due to the GTs' moving modes in the two scenarios being different, which influences the scheduling decisions a lot.

Recall that in the considered system setup, as shown in Fig. 2, the time length of the first stage as well as the whole operation time are actually influenced by the processing time of the last offloaded task. Hence, in our last simulation results, we study in Fig. 12(a) and (b) the relationship between the decision of the last offloaded task and the UAV's computation efficiency, while both the SA and LP scenarios are considered. From the figures, while increasing the CPU frequency of the UAV, we observe from both scenarios that the last offloaded task first changes, and then stays constant. For instance, when the frequency is less than 4 GHz in the scenario SA, S_1 is scheduled for the last offload task, and when the frequency is greater than 4 GHz, the last offload task scheduling is changed from S_1 to S_2 . In addition, it is observed that the data processing time gradually decreases when the frequency is greater than 4 GHz in both scenarios. These results actually indicate the trade-off between the offloading time, the processing time of the last offload task, and decision transmission time while minimizing the whole

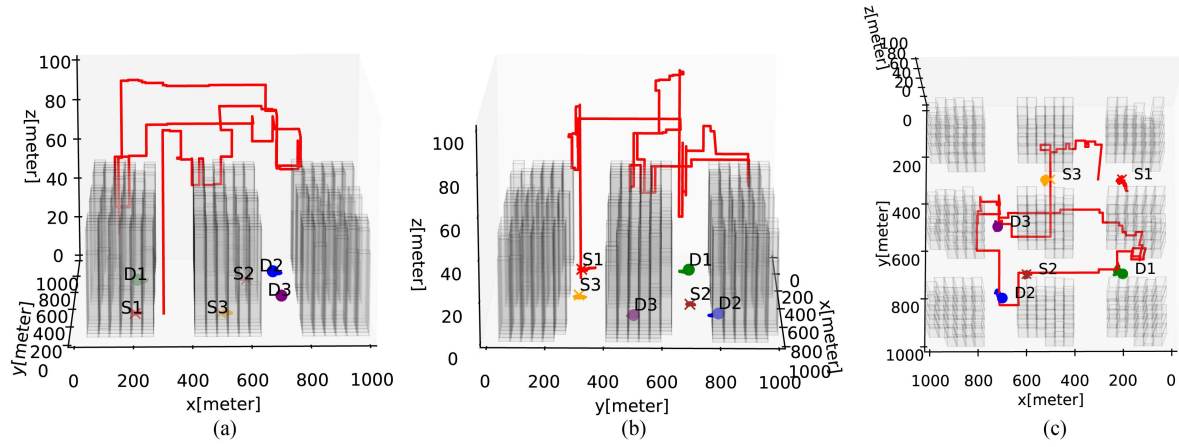


Fig. 9. Exhibition UAV's 3D flight trajectory as well as the movement trajectories of GTs in scenario SA. (a) Front view. (b) Left view. (c) Top view.

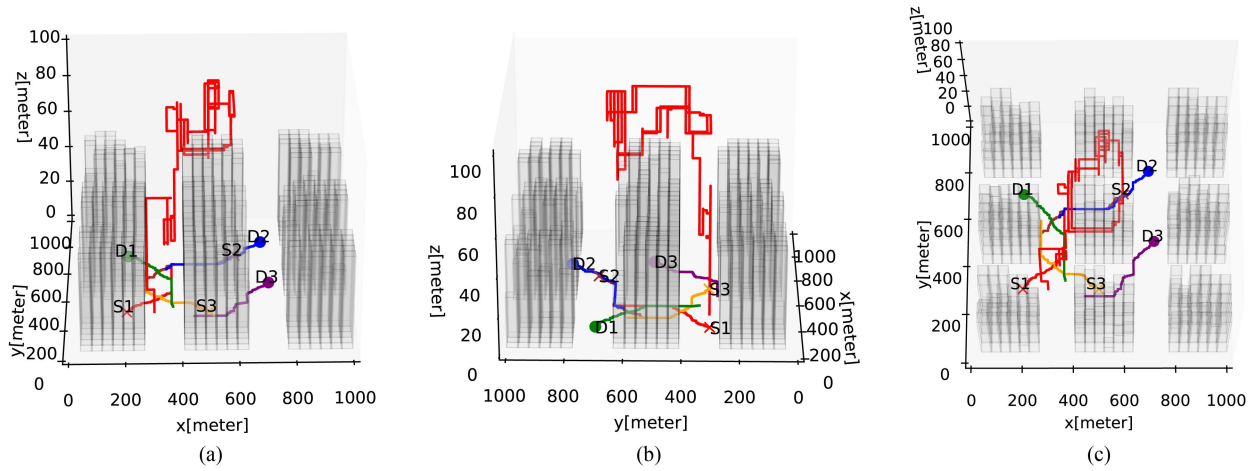


Fig. 10. Exhibition UAV's 3D flight trajectory as well as the movement trajectories of GTs in scenario LP. (a) Front view. (b) Left view. (c) Top view.

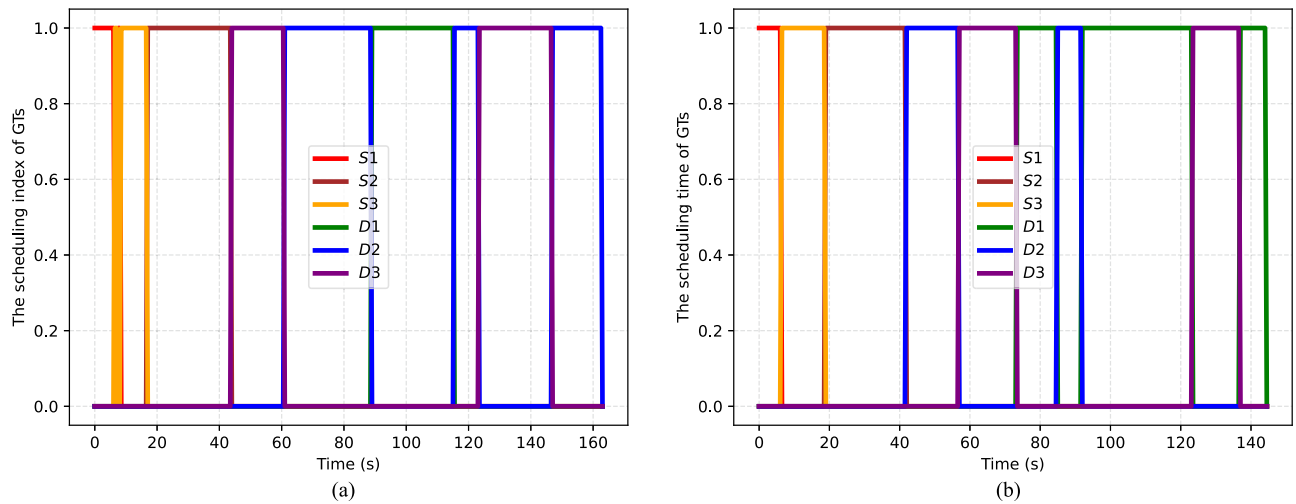


Fig. 11. Scheduling of GTs over the whole operation time under different scenarios. (a) Scenario of SA. (b) Scenario of LP.

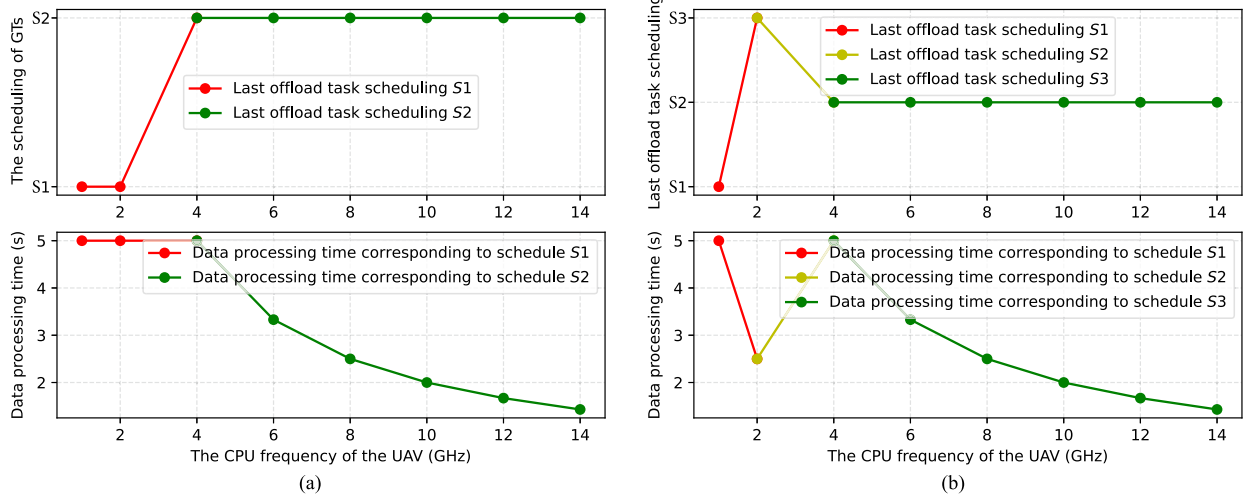


Fig. 12. GT's scheduling of the last offload task and the data processing time over the whole operation time under different scenarios. (a) Scenario of SA. (b) Scenario of LP.

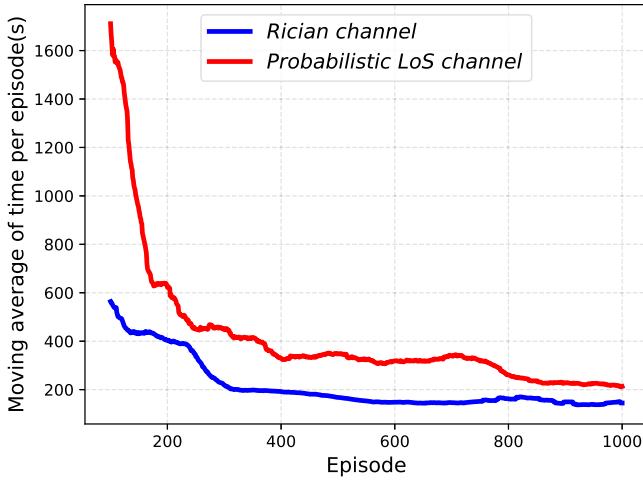


Fig. 13. Whole operation time under different channels in scenario SA.

operation time which is the sum of the three-time lengths. Increasing the CPU frequency actually reduces the influence of the processing time of the last offload task in such trade-off. When the CPU frequency is low, the processing time of the last offload task plays an important role, and the system prefers to select a light workload task as the last one. However, if the UAV's CPU is sufficiently powerful, the bottleneck of the system (in terms of minimizing the whole operation time) is not the processing time of the last offload task anymore. In such cases, keep increasing the CPU frequency will not change the decision.

Moreover, we evaluate the convergence performance of our proposed method under the Rician channel and compare its performance to that of the probabilistic LoS in scenario SA. The observation from Fig. 13 reveals the convergence of both the Rician channel and probabilistic LoS channel curves to a stable value. This demonstrates that our proposed method is also can be extended to the Rician channel scenario. In addition, as illustrated in Fig. 13, the UAV accomplishes the task in a shorter time under the Rician channel compared to the probabilistic LoS channel. This is primarily due to the different emphases during the modeling process of the two channel types. In the

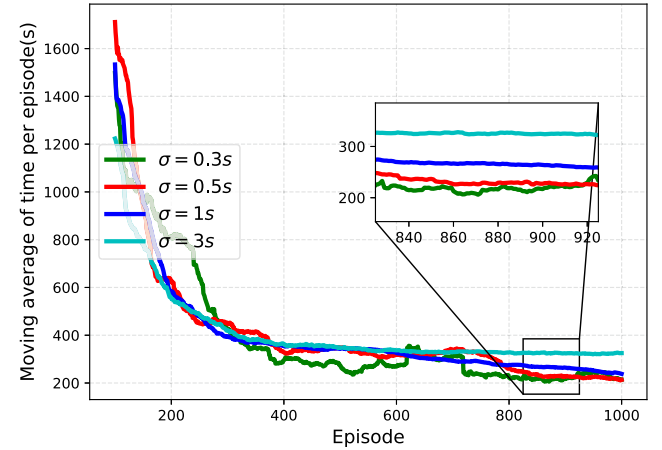


Fig. 14. Whole operation time under different σ in scenario SA.

modeling process, the Rician channel mainly focuses on the impact of signal reflection, refraction, and diffraction on signal transmission, while the probabilistic LoS channel is primarily concerned with the influence of shadow fading. In the scenario we consider, due to the low-altitude flight of the UAV, the channel transmission between the UAV and the GTs is predominantly affected by shadow fading. Thus, the probabilistic LoS channel is adopted in our work.

Finally, we investigate the influence of various σ selections on the completion time of UAV missions, as shown in Fig. 14. As can be seen from Fig. 14, increasing σ will accelerate the convergence speed of the algorithm, but the UAV mission completion time will increase. This confirms that the choice of σ is a trade-off between system performance and algorithm complexity. In addition, it can be seen from the figure that excessively small values of σ induce oscillations in the curve, i.e., the curve exhibits severe oscillations when $\sigma = 0.3s$. This occurs because as σ decreases, the algorithm's complexity increases, making the UAV more prone to getting trapped in a local optimal solution. Moreover, the proposed algorithm can converge well for different σ . This also verifies that the proposed algorithm has superior convergence properties.

VI. CONCLUSION

In this paper, we considered a UAV-assisted MEC network with multiple mobile GTs and probabilistic LoS channels. We provided a design minimizing the whole operation time via jointly optimizing the UAV 3D trajectory together with the scheduling of offloading transmissions from source GTs and decision transmissions to destination GTs. In particular, an unrestricted feasible domain for the UAV flight is adopted in the design while obstacle avoidance to the 3D building is guaranteed. To address the problem, we first transform the established problem into an MDP and then propose a D3QN-based method in which the UAV acts as an agent to explore and learn the policy of 3D trajectory and scheduling GTs. Via simulations, the convergence of the proposed method is confirmed, and appropriate hyperparameters of the proposed method are also investigated. In addition, we investigated the performance of the proposed method in terms of operation time and robustness, as well as the impacts on the system setups, e.g., GTs' moving model, UAV flying altitude, on the designed 3D trajectory, and scheduled results.

We finalize the whole paper by spotlighting its high extensibility. On the one hand, this paper is the first endeavor to propose a joint optimization of 3D UAV trajectory and GTs' communication scheduling by taking into account various practical aspects of the scenarios, including 3D buildings obstacle avoidance, GTs with mobile capability, and feasible UAV flying zone among buildings. Through simulation experiments, it was discovered that the proposed method has a strong robustness performance under different setups with respect to different GTs mobility behavior scenarios, different initial conditions, various UAV flying altitude ranges, and trajectory designs in different dimensions. In addition, allowing UAV to fly at a low altitude can significantly reduce mission completion time. Furthermore, compared to 2D trajectory design, the performance improvement achieved with 3D trajectory design is mainly attributed to its increased design freedom in both transmission distance and elevation angles. Moreover, the proposed method can be extended to other system designs, e.g., joint resource allocation involving transmitting bandwidth and computation frequency, designs that allow GTs to fly, multi-UAV joint trajectory designs, other practical channels, etc. On the other hand, in order to solve the joint optimization problem, a modified DDQN, i.e., the D3QN method is introduced, which accelerated the convergence speed and improved the stability of the method. The proposed D3QN method is expected to facilitate the designs for scenarios having non-convex and complicated constraints and difficulties achieving convergence.

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